# 4. Alert Triage with Automation

## 1. Objective

To implement an automated SOC alert triage workflow using Wazuh SIEM for detection, VirusTotal for threat intelligence validation, and TheHive for incident management and case tracking.

## 2. Tools Used

- Wazuh SIEM (Log monitoring and alert generation)
- VirusTotal (Threat intelligence and reputation analysis)
- TheHive (Incident response and case management)

## 3. Triage Simulation in Wazuh

A simulated security event representing a **"Suspicious File Download"** was generated in the Ubuntu endpoint. Wazuh SIEM detected this activity based on predefined log monitoring rules and generated a high-severity alert.
The alert was reviewed in the Wazuh dashboard and key details were extracted for further investigation.

**Alert Details**

| Alert ID | Description | Source IP | Priority | Status |
|---|---|---|---|---|
| 005 | File Download | 192.168.1.102 | High | Open |

## 4. Alert Analysis Process

The alert was analyzed through the following SOC triage steps:

- Detection of suspicious file download activity
- Extraction of source IP and event metadata
- Classification as a high-priority security event
- Marking alert as **Open** for further validation

This represents the initial **Level 1 SOC analyst triage phase**.

## 5. Threat Intelligence Validation (VirusTotal)

The extracted indicator (http://192.168.1.102/) was analyzed using VirusTotal.

**Results:**

- Detection ratio: **0 / multiple security vendors**
- No malware or phishing signatures detected
- All listed vendors marked the URL/IP as **Clean**
- Classified as **private/internal network activity**

**Interpretation:**

Although VirusTotal showed no malicious detection, SOC analysis still treats this as a **contextual security event**, since the activity originated from a simulated suspicious file download in Wazuh. Correlation with SIEM logs is required before final classification.

📌 *VirusTotal result screenshot should be attached here as evidence.*

**6. Incident Management using TheHive**

The alert was escalated to TheHive for structured incident response.

**Case Details:**

- Case Title: Suspicious File Download Alert
- Severity: High
- Status: Open
- Linked Artifacts:
  - Wazuh SIEM alert logs
  - VirusTotal analysis results

TheHive enables centralized tracking of the incident lifecycle from detection to resolution.

**7. Automated SOC Workflow**

The implemented SOC triage pipeline is as follows:

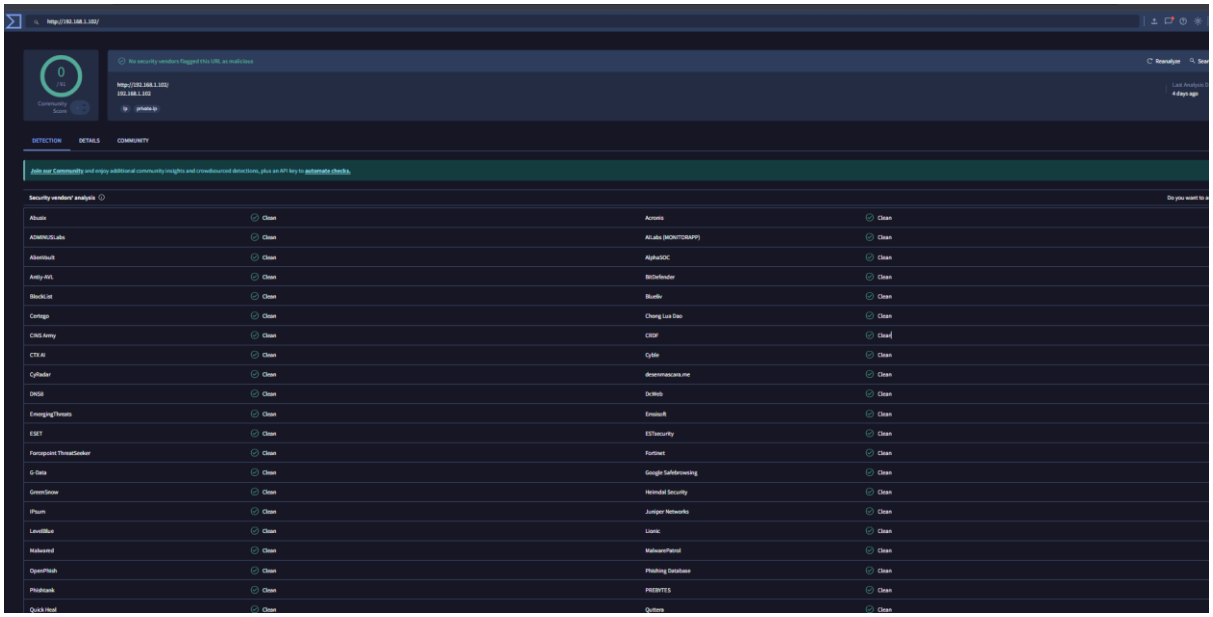
Wazuh Alert Generation
↓
Log Detection & Alert Creation
↓
Indicator Extraction (IP / URL)
↓
VirusTotal Threat Intelligence Check
↓
TheHive Case Creation & Tracking
↓
Analyst Decision (Investigate / Monitor / Close)

**8. Results and Observations**

- Wazuh successfully generated and detected simulated suspicious activity
- Alert was correctly classified as high severity
- VirusTotal confirmed no malicious vendor detections
- TheHive successfully recorded and managed the incident
- End-to-end SOC triage workflow was successfully implemented

**9. Conclusion**

A semi-automated SOC alert triage system was successfully implemented using Wazuh, VirusTotal, and TheHive. The system demonstrated real-world SOC operations including detection, threat intelligence validation, and incident management. This workflow improves analyst efficiency by reducing false positives and enabling structured security response.


http://192.168.1.102/
No security vendors flagged this URL as malicious
Reanalyze
0
/ 95
Community Score
http://192.168.1.102/
192.168.1.102
ip
private ip
Last Analysis Date
4 days ago
DETECTION
DETAILS
COMMUNITY
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to a
Abusix Clean
Acronis Clean
ADMINUSLabs Clean
AILabs (MONITORAPP) Clean
Abertvault Clean
AlphaSOC Clean
Antiy-AVL Clean
BitDefender Clean
BlockList Clean
Bluelliv Clean
Certego Clean
Chong Lua Dao Clean
CINS Army Clean
CRDF Clean
CTI.AI Clean
Cyble Clean
CyRadar Clean
desenmascara.me Clean
DNSB Clean
DrWeb Clean
EmergingThreats Clean
Emsisoft Clean
ESET Clean
ESTsecurity Clean
Forcepoint ThreatSeeker Clean
Fortinet Clean
G-Data Clean
Google Safebrowsing Clean
GreenSnow Clean
Heimdal Security Clean
IPsum Clean
Juniper Networks Clean
LevelBlue Clean
Lionic Clean
Malwared Clean
MalwarePatrol Clean
OpenPhish Clean
Phishing Database Clean
Phishtank Clean
PREBITES Clean
Quick Heal Clean
Quttera Clean